Validation of a method for the detection of virulent Yersinia enterocolitica and their distribution in slaughter pigs from conventional and alterna...

Various methods have been described in the literature for the detection of virulent Yersinia enterocolitica in pigs. The risk factors for pig herd contamination have yet to be determined. The objective of this study was to validate a sensitive method for the detection of Y. enterocolitica and to describe the distribution of the bacteria in pigs at slaughter from conventional and alternative ("organic") housing systems. First, samples were collected from tonsils, caecum with caecal contents, and the caecal lymph nodes of 60 slaughter pigs. These samples were used to compare the sensitivity of six different laboratory culture methods either in common use or described in the literature with that of a polymerase chain reaction with two primer pairs (multiplex PCR). Then, only PCR was used to examine tonsils, caecum and caecal lymph nodes from two groups of slaughter pigs: 210 from six conventional fattening farms and 200 from three with alternative housing. The results of the multiplex PCR were positive in 28 cases. All culture methods proved inferior to PCR in sensitivity. In the second part of the study, PCR detected 36 (18%) positive pigs from alternative housing and 60 (29%) from conventional housing ($p<0.05$). The highest rate of Y. enterocolitica contamination was found in tonsils (11% alternative, 22% conventional; $p<0.05$), followed by caecum (5%, 11%) and lymph nodes (2%, 7%). The housing system appears to be one important factor in the prevalence of this common pathogen in pig herds, as we found important differences between the two systems studied here. In the conventional system, the main risk factors appeared to be sourcing pigs from different pig suppliers, use of commercial feed and transportation to slaughter.